

counting for an estimated 70-80% of electronic components and devices such as laptops, while semiconductors remain around 90-95% import-dependent, as per Niti Aayog.

To bridge these gaps, the government did roll out production-linked incentive (PLI) schemes for electronics manufacturing and component production, alongside the \$10-billion Semicon India programme, launched in 2022. Despite the policy push and growing investments, the journey towards a self-sustaining electronics ecosystem remains a work in progress. There are hurdles of high capital costs, infrastructure gaps, talent shortages, technology dependence, and supply chain constraints.

THE AI ERA

The advent of AI has amplified these vulnerabilities. Velamakanni argues that as a capital-scarce, labour-rich country, India naturally gravitated towards services. However, with a more mature venture capital and private equity ecosystem, a growing pool of risk-taking entrepreneurs, and falling costs of product development, it is better positioned to build globally competitive technology products and intellectual property (IP).

India is also rapidly expanding its data centre infrastructure. Investments are accelerating. Google is reportedly planning a \$15-billion investment to develop a one-gigawatt AI-focused data centre campus in Visakhapatnam, Andhra Pradesh. More recently, Reliance Industries and Meta have agreed to collaborate on building an AI data centre.

TECH SOVEREIGNTY

India remains almost entirely dependent on NVIDIA GPUs for advanced AI computing, though consultants note that this is not unique to India but a reality across the



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GROUP CHIEF
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AND CHAIRPERSON,
NASSCOM



“Beyond the most advanced semiconductor nodes such as 5 nanometre and 3 nanometre, India will have to develop capabilities across the entire value chain”

PANKAJ MOHINDROO
CHAIRMAN, ICEA

world. According to Peeyush Vaish, Partner and TMT Industry Leader at Deloitte India, no country today possesses complete control over the entire technology value chain.

For India, semiconductors remain a strategic priority. “As and when you get into more automation, the demand for semiconductors in India will significantly go up over the next five years. It will probably go up by 10x or even 20x,” says Vaish.

Yet, building semiconductor capability alone may not be sufficient. Experts argue that India must also become deeply embedded in global manufacturing networks if it hopes to move up the technology value chain.

Mohindroo points to another structural challenge—India’s limited integration into global value chains (GVCs). “One big structural defect India has is that GVCs don’t have major participation in India, barring Apple,” he says.

Beyond manufacturing and supply chains lies a broader debate around what technological sovereignty should mean in the AI era. For Velamakanni, the ultimate test of tech sovereignty is control.

While experts differ on how much self-reliance is realistically achievable, they agree that technological sovereignty is not about eliminating every dependency. Rather, it is about ensuring that critical technologies, infrastructure and supply chains do not become strategic vulnerabilities.

India’s challenge is no longer proving that it can assemble the world’s devices or provide software talent to global companies. It is ensuring that the critical technologies underpinning the digital economy cannot be switched off by someone else. **BT**

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